



Mechanism Exploration of Environmental Pollutants on Premature Ovarian Insufficiency: a Systematic Review and Meta-analysis

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Received: 10 May 2023 / Accepted: 7 August 2023 / Published online: 23 August 2023
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Abstract

As a public health problem, premature ovarian insufficiency leads to infertility or sub-fertility. In addition to premature ovarian insufficiency (POI) increases the lifetime risk of bone fragility, cardiovascular disease, and cognitive impairment. To investigate the effects of environmental pollutants on the occurrence of POI and explore its mechanism, we conducted a computer search for articles published in electronic databases by December 13, 2022. Three reviewers independently examined all included studies and scored the qualities of included studies using the Newcastle-Ottawa Scale criteria. In this meta-analysis, eight clinical studies as well as ten preclinical findings showed a pooled OR of 2.331 and 95% CI of 1.968–2.760. This confirms that environmental pollutants, including POPs, heavy metals, PAEs, PAHs, cosmetic and pharmaceutical products, and cigarette smoke, are indeed significant risk factors for POI. In addition, it is demonstrated from the results of this study that signaling pathway of calcium and PI3K Akt and *Xpnpep2*, *Col1*, *Col3*, *Col4*, *Cx43*, *Egr3*, *Tff1*, and *Ptgs2* genes may all be involved in the process. Environmental pollutants, including POPs, heavy metals, PAEs, PAHs, cosmetic and pharmaceutical products, and cigarette smoke, are indeed significant risk factors for POI.

Keywords Environmental pollutant · Mechanism · Meta-analysis · Premature ovarian failure · Premature ovarian insufficiency

Introduction

Premature ovarian insufficiency (POI) refers to impaired ovarian function in women before the age of 40 [1], characterized by amenorrhea or oligomenorrhea for at least 4 months, and an increased FSH level (> 25 IU/I) on two occasions 4 weeks apart [2]. The prevalence of POI varies among studies and it depends on population characteristics, such as racial, economic, and social factors. According to the study of Women's Health Across the Nation, 1.4% of Hispanic, 1.4% of African American, 1.4% of African American, 0.5% of Chinese, and 0.1% of Japanese females experienced POI. POI was more prevalent in countries with low and medium human development indices. POI is a public health concern because it results in sub-fertility or infertility [3]. Moreover, we have observed a distinct negative correlation between premature ovarian insufficiency and female reproductive well-being. Premature ovarian insufficiency results in a reduction in the quantity of oocytes in women, as well as disruptions in cyclic ovulation, thereby increasing the risk of infertility [1]. Additionally, premature

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ovarian insufficiency can lead to early menopause and a decline in estrogen levels, potentially giving rise to women's health concerns such as decreased bone density and osteoporosis [4]. It may even entail a multitude of health risks, including cardiovascular diseases, breast cancer susceptibility, and mental health issues [5, 6].

The causes of POI are still poorly defined, since in more than 75% of cases the etiology is not known [7, 8]. It has been reported that genetics [9], metabolic [10], infectious [11], and immunologic [12] factors may be related with POI. Environmental pollutants seem to be important factors influencing the ovarian function. There are many processes that can lead to the release of environmental toxicants, including manufacturing, combustion, leaching from products, and human contamination [13]. For instance, cigarette smoke contains a variety of chemicals. Approximately 7000 chemicals are released in cigarettes when they are burned, including aromatic amines, aldehydes, phenols, aromatic hydrocarbons, and heavy metals [14]. Compared to non-smokers, current smokers have elevated levels of FSH (up to 23%) and lower levels of serum estradiol [15, 16]. Moreover, smoking has been associated with decreased serum concentrations of inhibin B and AMH, which are indicative of ovarian aging [17, 18]. Polycyclic aromatic hydrocarbons (PAH) are also ubiquitous pollutants that accumulate in the environment, cigarette smoke, and even food [19]. It has been reported that women with elevated serum PAH levels suffer from polycystic ovary syndrome, abnormal sex hormone levels, and POI [20, 21].

In this study, we conducted a systematic review and meta-analysis to investigate the effects of environmental pollutants on the occurrence of POI and explore its mechanism.

Materials and Methods

The study is a meta-analysis which contains no participants and ethics approval is not applicable. The meta-analysis was registered with PROSPERO on January 27, 2023 (CRD42023392256).

Search Strategy

We conducted a computer search for articles published in electronic databases (PubMed, Embase, Web of Science, and The Cochrane Library) by December 13, 2022. The following combined keywords were used. ("premature ovarian failure" or POF) and ("environmental pollutants" or "endocrine disruptors*" or Eds or pesticides* or "polyaromatic hydrocarbons" or phenol or bisphenol or "flame retardants" or phthalates* or dioxins or phytoestrogens or tobacco or smoke or cigarettes or cosmetics or xenobiotics or "polychlorinated biphenyls*" or PCBs or β -hexachlorocyclohexane

or mirex or "p'-DDE" or furans or "PAHs" or PAHs or calcium or selenium or "vitamin D" or zinc or copper or dichlorodiphenyltrichloroethane or DDT or "perfluorooctane sulfonate" or PFOS or diethylstilbestrol or DES or triclosan or phytoestrogens or parabens or "bisphenol A" or BPA or genistein).

Inclusion and Exclusion Criteria

Inclusion criteria were (1) patients with premature ovarian failure (POF) or primary ovarian insufficiency (POI), (2) adult women older than 18, (3) environmental pollutants, (4) type of study is observational, and (5) were published in English. Exclusion criteria were (1) abstracts, letters, talks, or reviews, (2) patients with non-premature ovarian failure or primary ovarian insufficiency (POI), and (3) include duplicate studies with data already included in the meta-analysis.

Data Extraction

Three reviewers independently examined all included studies and extracted the following data: author; publication year; type of study design; preclinical or clinical study; model or patient details, region of data set, type of pollutants; details of environmental pollutants; indicator; results; qualitative and quantitative results; other exposure; objects; and diagnostic evaluation. Differences of opinion were resolved by consensus among the three independent reviewers.

Quality Assessment

Three reviewers independently scored the qualities of included studies using the Newcastle-Ottawa Scale (NOS) criteria. According to the NOS, which can give a maximum score of nine. A "star system" is judged on three broad perspectives: the selection of the study groups; the comparability of the groups; and the ascertainment of either the exposure or outcome of interest for case-control or cohort studies respectively.

Statistical Analysis Methods

The Cochran Q test and I^2 statistics were employed to analyze study heterogeneity, and I^2 was calculated using the equation: $I^2 = 100\% (Q \text{ df})/Q$. I -square less than 50% was considered as low heterogeneity between studies and greater than 50% was considered as high heterogeneity. If heterogeneity was small or absent (I^2 less than 25%), a fixed-effects model was chosen. If heterogeneity was large (I^2 greater than 50%), a random effects model was selected. p less than 0.05 indicates statistical significance, and all statistical analyses were completed by STATA 15.1.

Results

Basic Characteristics

The flowchart for the selection of eligible studies is showed in Fig. 1. A total of 1131 publications were identified through database searches. The detailed search strategy is in Online Resource 1. After eliminating duplicate records, the titles and abstracts of the remaining 912 publications were reviewed. After review, 84 of these publications were eligible. After reviewing the full text of 84 articles, 66 of these publications did not meet our inclusion criteria: 5 were letters, 1 exposure was not an environmental contaminant, 17 had inconsistent endings, 5 had no quantitative data, 1 had inconsistent language, and 37 had inconsistent themes. Finally, 18 articles were determined to be eligible for inclusion in this meta-analysis.

The Newcastle-Ottawa Scale (NOS) scores of the included studies are shown in Online Resource 2. Most of

the studies were of good quality, with quality assessment scores of 7 or more (out of 9).

Meta-analysis Results

Table 1 provides an overview of the characteristics of the included studies. In total, there were 10 different environmental pollutants, all of which were studied in adult females, and a total of 7 case-control studies. The study involved a total of 1321 patients in the case group and 1662 patients in the control group.

The results are shown in Fig. 2 and the pooled OR was 2.331 with a 95% CI of 1.968–2.760. The final analysis was $p=0.0001<0.05$, indicating that the results were statistically significant. Environmental pollutants are harmful factors for ovaries and environmental pollutants may cause premature ovarian failure.

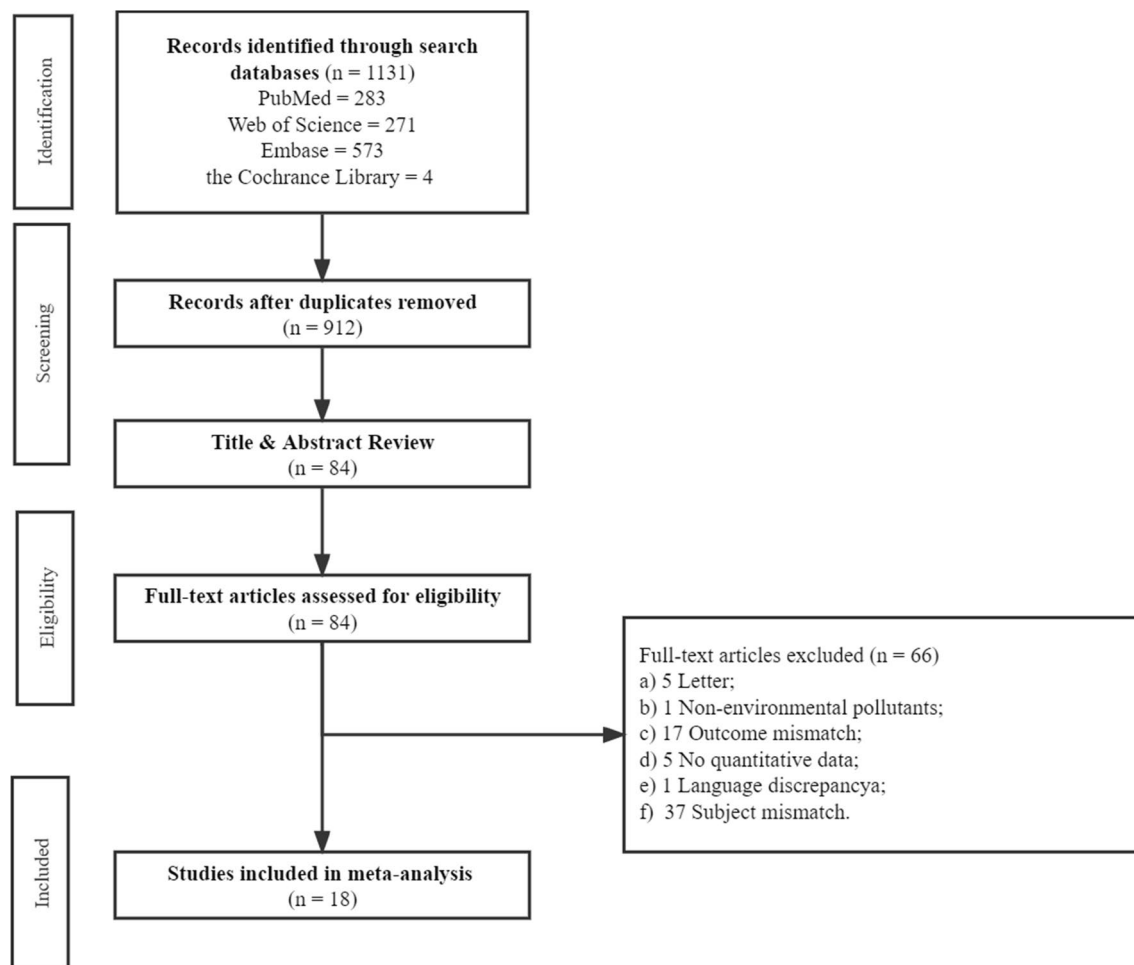
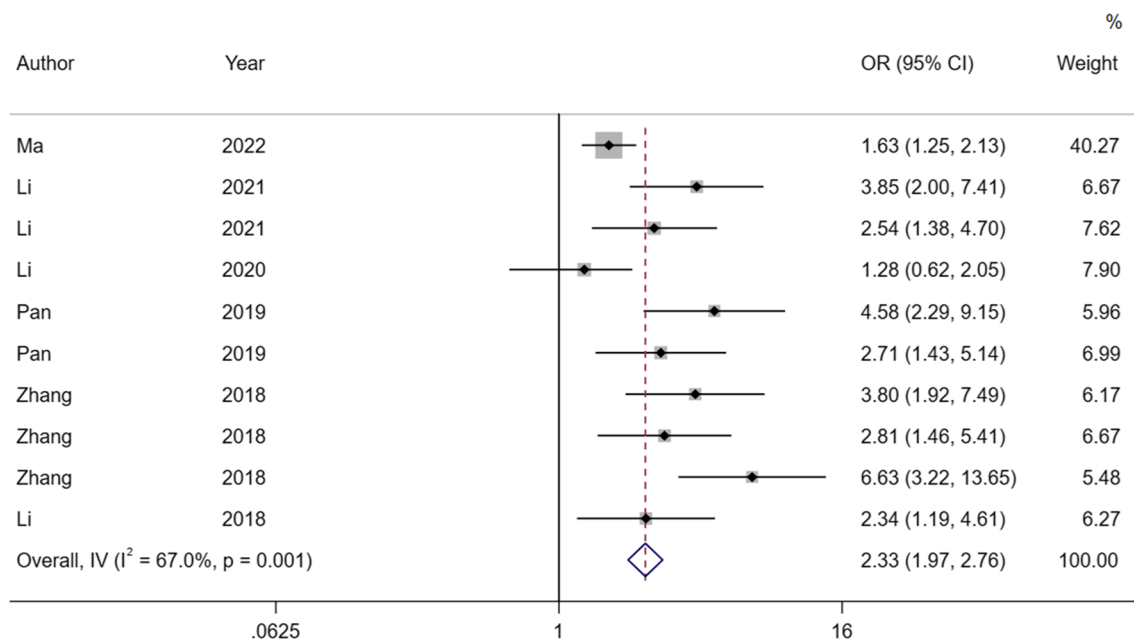


Fig. 1 Study flowchart of selection of eligible studies

Table 1 Characteristics of the included studies

Author	Year	Study design	Region	Patients	Environmental pollutants	Objects	N (case group)	N (control group)
Ma	2022	Case-control study	China	Adult	Thallium	Ovary	169	209
Li	2021	Case-control study	China	Adult	Cu	Ovary	169	209
Li	2021	Case-control study	China	Adult	Se	Ovary	169	209
Li	2020	Case-control study	China	Adult	Bisphenol A (BPA)	Ovary	/	/
Pan	2019	Case-control study	China	Adult	PCB	Ovary	141	214
Pan	2019	Case-control study	China	Adult	p,p'-DDTs	Ovary	141	214
Zhang	2018	Unknown	China	Adult	PFOA	Ovary	120	120
Zhang	2018	Unknown	China	Adult	PFOS	Ovary	120	120
Zhang	2018	Unknown	China	Adult	PFHxS	Ovary	120	120
Li	2018	Case-control study	China	Adult	Pyrethroids	Ovary	172	247

**Fig. 2** The forest plot shows the aggregated results

Characteristics of the Preclinical Studies

Table 2 [22–31] provides an overview of the characteristics of the included preclinical studies. Of the 10 articles included in this systemic review, seven ones reported models of rat, and the other three mouse. Most studies described the influences of industrial additive on POF/POI, including phthalic acid esters (PAEs) [26, 29] and polycyclic aromatic hydrocarbons (PAHs) [23, 24, 28]. Some showed another three types of environmental pollutants, such as cosmetic and pharmaceutical products [25, 27], cigarette smoke [22, 23], and heavy metals [30, 31]. As could be seen from the summary of key results from these preclinical studies,

environmental pollutants would lead to POF through different mechanistic pathways, in a performance of improper functioning in the hypothalamic-pituitary-gonadal (HPG) axis, changed mRNA or serum hormone level in follicles, reduced ovarian volume, and obvious follicle atresia.

Discussion

The majority of research focused on the effects of industrial additives on POF/POI, such as PAEs and PAHs. According to the summary of key findings from these preclinical

Table 2 Characteristics of the preclinical studies

Author	Year	Model/species	Environmental pollutants	Objects	Key results
da Costa [31]	2021	Rat	Cadmium	Ovary	Displayed metabolic impairments, such as a reduction in adiposity, dyslipidemia, and insulin resistance; caused improper functioning in the hypothalamic-pituitary-gonadal axis
Lee [27]	2017	Rat	Parabens	Ovary	Increased the <i>Amh</i> mRNA level but unchanged <i>Foxl2</i> and <i>Kitlg</i> acting in primordial follicles; induced an increase in FSH levels in serum and significantly decreased the total number of follicles
Hong [25]	2018	Rat	TiO ₂ NP	Ovary	Resulted in premature ovarian failure, reductions in the levels of estradiol, progesterone, and inhibin B, and increases in luteinizing hormone, FSH, FSH/LH ratio, AMH, TSH, free triiodothyronine, free tetraiodothyronine, anti-nuclear antibody, and anti-thyroid peroxidase antibody levels in serum
Tran [26]	2018	Rat	VCD and phthalate, including DEHP, BBP, and DBP	Ovary	Significantly reduced the numbers of primary follicles; significantly decreased the expressions of <i>Amh</i> and <i>Sohlh2</i> ; significantly increased serum levels of follicle-stimulating hormone
Li [22]	2022	Mouse	Cigarette smoke	Ovary	Decreased the total ovarian volume; the ovaries had fewer follicles at each developmental stage (primordial, primary, secondary, and antral follicles); caused a prolonged estrous cycle; increased serum FSH and decreased AMH and E2 levels
Guo [28]	2022	Rat	Phenanthrene	Ovary, follicle	Follicular atresia significantly exacerbated (1.85-fold)
Banu [30]	2015	Rat	CrVI	Ovary, follicle	Accelerated germ cell nest breakdown and advanced primordial and primary follicle formation; decreased healthy primordial and primary follicles and increased follicle atresia
Zhang [29]	2015	Mouse	DEHP	Follicle	Significantly decreased the number of primordial follicles; increased the number of secondary follicles
Siddique [23]	2014	Mouse	CSC and B(a)P	Follicle	A dose-related increase in 8-IsoP concentrations; significantly higher 8-OH-dG
Rhon-Calderon [24]	2018	Rat	3MC	Follicle	18 of the 29 genes studied were up-regulated, five were down-regulated and six were not altered

FSH follicle-stimulating hormone, *TiO₂ NP* titanium dioxide nanoparticles, *LH* luteinizing hormone, *AMH* anti-Müllerian hormone, *TSH* thyroid-stimulating hormone, *VCD* 4-vinylcyclohexene diepoxide, *DEHP* di-2-ethylhexyl phthalate, *BBP* butyl benzyl phthalate, *DBP* dibutyl phthalate, *CrVI* hexavalent chromium, *CSC* cigarette smoke condensate, *B(a)P* benzo(a)pyrene, *8-IsoP* 8-isoprostane, *8-OH-dG* 8-hydroxy-2-deoxy Guanosine, *3MC*, 3-methylcholanthrene

studies, environmental pollutants cause POI via various mechanistic pathways.

In the included clinical studies, the effects of multiple environmental pollutants on POI were reported, including persistent organic pollutants (POP) [32–35], heavy metals [36, 37], PAEs [38], and PAHs [21]. The findings illustrated that most of the environmental pollutants, including thallium, Cu, Se, polychlorinated biphenyls (PCB), p,p'-dichlorodiphenyltrichloroethane (p,p'-DDT), perfluorooctanate (PFOA), perfluorooctane sulfonate (PFOS), perfluorohexanesulfonate (PFHxS), and pyrethroids, except for BPA, were all shown to be significant risk factors for POI. Li et al. [33] suggested that BPA exposure at a low level was not linked to POI in Chinese women (OR = 1.28, 95% CI, 0.62–2.05). This speculation was validated in similar

findings, which discovered that although blood levels of BPA and phthalate metabolites were elevated in women with POI, mono-buthyl phthalate (MBP), on the other hand, was more likely a risk factor in the development of POI [39]. In addition, PFHxS had the highest OR as a risk factor (6.63; 95% CI, 3.22–13.65) among all the environmental pollutants, probably because as a persistent synthetic chemical, PFHxS was widely used in industry and consumer production [34]. PFHxS also caused hypothyroidism by affecting triiodothyronine (T3) and thyroxine (T4), thereby manifesting hyperprolactinemia and further contributing to female reproductive impairment [40–42].

Results of preclinical studies in rats or mice showed morphological or histological results that environmental pollutants would reduce the numbers of follicles at different

developmental stages (primordial, primary, secondary, and antral follicles) and total ovarian [25–27, 29]. Besides, follicular atresia is significantly exacerbated [28, 30]. Furthermore, AMH levels were drastically reduced in animal models exposed to environmental contaminants [22, 26, 27]. There have also been multiple instances of POI follicles growing to the antral stage, in which *Amh* and its mutant gene may be strong predictors of the existence of follicles, and lower AMH immunoexpression in antral follicles might indicate a failure in antral development [43, 44]. Nevertheless, one of these studies showed a TiO_2 -induced increase in AMH levels in serum combined with a decrease in inhibin B levels, which suggests this contaminant exposure may be closely associated with the generation of POI in mice [25]. Likewise, although less sensitive than serum AMH levels, FSH levels are routinely used to diagnose POF [45], and higher serum FSH levels induced by environmental pollutants were detected in the animal models [22, 25–27]. Moreover, heavy metal exposure, such as cadmium, would produce metabolic deficits such as decreased adiposity, dyslipidemia, and insulin resistance, as well as incorrect hypothalamic-pituitary-gonadal axis functioning [31], which was in line with earlier ovarian research [46].

Because only 25% of POI patients have an identified cause [47], there are no consistent conclusions regarding the mechanism by which environmental pollution leads to POI. However, it is evident from the results of this study that the signaling pathway of calcium and PI3K-Akt [28]; and *Xpnpep2*, *Col1*, *Col3*, *Col4* [30], *Cx43*, *Egr3*, *Tff1*, and *Ptgs2* [29] genes may all be involved in the process. Besides the mechanism at the molecular level, leading to a decline in follicular development and premature ovarian failure by stimulating oxidative stress was also one of the hypotheses [23]. Although the processes by which various environmental contaminants generate POI may differ, the currently known mechanisms all revolve around these four: (a) decrease of the resting primordial follicle pool (due to assembly failure or acquired owing to enhanced atresia), (b) increased follicular atresia, (c) increased primordial follicle activation, and (d) ovulation blocking by follicles occurring before the antral stage. Therefore, in addition to the mechanism of inducing oxidative stress mentioned above, environmental pollutants can also occur through endocrine-disrupting chemicals as well as epigenetic modifications [47].

Although the underlying mechanisms of pollutant impact on POI are still inconclusive, numerous studies have confirmed its adverse clinical effects, affecting the normal development and function of the ovaries [48, 49], leading to ovarian tissue damage and functional decline [47]. The potential influence of pollutants on premature ovarian insufficiency underscores the heightened importance of nutrition and health management. It is imperative to implement preventive strategies from various perspectives, including patient care. Within

the scope of nursing, attention should be given to patients' dietary and lifestyle choices, such as increasing the intake of foods rich in antioxidants and nutrients while minimizing exposure to environmental pollutants [50, 51]. Enhancing patient education on the risks and consequences of pollutant exposure is crucial. Healthcare providers can offer information on environmental protection, pollutant avoidance, and reduction of exposure to assist patients in taking proactive measures to safeguard their health [52]. Encouraging regular check-ups and monitoring, as well as adherence to physicians' treatment recommendations, is also essential. Regular gynecological examinations and relevant tests can aid in the early detection of ovarian dysfunction, allowing for appropriate interventions to mitigate the impact of pollutants on premature ovarian insufficiency [53]. While the study did not restrict its scope to a specific region, it solely included the Chinese population, potentially introducing geographical bias to the results. Additionally, research suggests interactions between environmental factors and individual genetics, lifestyle, and other health factors [54], which may contribute to potential biases or confounding factors. Despite supporting research on underlying mechanisms, the current clinical evidence remains limited, necessitating further population studies, long-term tracking research, and generalization of results across regions, including Asian and even global population epidemiological data and laboratory research, to fill the existing gaps in knowledge [55–57].

Conclusions

Environmental pollutants pose a serious threat to human and animal reproduction, and their deleterious effects can perturb endocrine and reproductive functions. It is clear that environmental pollutants, including POPs, heavy metals, PAEs, PAHs, cosmetic and pharmaceutical products, and cigarette smoke, are indeed significant risk factors for POI, with pooled OR of 2.331. And preclinical studies speculated that environmental pollutants lead to POF via improper HPG axis functioning, changed follicular mRNA/hormones, reduced ovarian volume, and obvious follicle atresia.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s43032-023-01326-5>.

Author Contribution Xiaodan Zhu contributed to conceptualization, data curation, investigation, drafting of the manuscript, and reviewing and editing of the manuscript. Meixia Liu contributed to statistic analysis, drafting of the manuscript, and reviewing and editing of the manuscript. Ruoxi Dong contributed to drafting of the manuscript, and reviewing and editing of the manuscript. Liqun Gao, Jiazhen Hu, Xinpei Zhang, and Xiaomei Wu contributed to conceptualization, data curation, investigation, and methodology. Bozhen Fan, Chao Chen, and Wenjuan Xu contributed to project administration and supervision. All authors have read and approved the final manuscript.

Funding This work was supported by the Xinglin Scholars Project of Chengdu University of Traditional Chinese Medicine [grant number YYZX2022169] and the Scientific Research Plan Project of Shanghai Science and Technology Commission [grant number 18411970500].

Data Availability The data underlying this article are available in the article and in its online supplementary material.

Declarations

Ethics Approval The study is a meta-analysis which contains no participants and ethics approval is not applicable.

Consent to Participate Not applicable.

Competing Interests All authors declare no competing interests.

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